Live AR Sudoku Solver

Goal. Point a webcam at a printed Sudoku. The app detects the grid, reads the digits, solves the puzzle and overlays the solution on the paper in real time, following the paper as it moves.

Pipeline.

1. Grid detection: adaptive threshold, Canny edges, largest 4-sided contour, and Hough lines to refine the grid.
2. Rectification: homography from the 4 corners to a flat square image.
3. Cell reading: split into 81 cells, detect empty cells, classify digits with a small CNN trained on MNIST plus printed-font digits.
4. Solving: backtracking with constraint propagation.
5. AR overlay: draw the solution through the inverse homography and track the corners across frames instead of re-detecting every frame.

Beyond existing hobby versions. Simple versions exist online, so we focus on:

- robustness to tilt, glare, shadows and curved paper;
- a stable overlay from frame to frame;
- a measured comparison of contour-based vs Hough-based grid detection.

Evaluation. About 50 photos and 10 videos under varied conditions. Metrics: grid detection rate, digit accuracy per cell, fully-solved rate, FPS, overlay jitter.

Course fit. Edges, Hough transform, thresholding and segmentation, homography, recognition.

Risk. Low: every stage has a known baseline, so effort goes into robustness and the demo.